VeriVote: Hybrid Electronic Voting System

Problem Statement

Kenya's electoral system faces major legitimacy crisis mainly in electoral violence and coercion, rigging/fraud allegations, and constitutional mandate without implementation.

Kenya's Majoritarian electoral system creates high stakes contests that have historically resulted in devastating violence. For instance, the 2007-2008 post-election violence caused over 1100 deaths, 3,500 injuries, and displaced around 600,000 people (International Criminal Court, 2010). Notably, deviation from electoral rules often stems from bribery or coercion. Shockingly, some voters even justify electoral malpractice as a necessary measure to counter perceived rigging by opponents (National Crime Research Center, 2016). Section 10 of Kenya's Elections Offences Act (2016) specifically addresses undue influence through coercion, threats, or intimidation that obstructs voters' freedom to exercise their electoral rights. However, the enforcement of the Election Offences Act still remains inadequate. Consequently, the coercion problem manifests forms such as vote buying during campaigns, post-election violence when results are disputed, and though rarely, intimidation at polling stations. As such, traditional paper-based voting offers no mechanism for voters to signal distress or change their vote after coercion, leaving vulnerable populations effectively disenfranchised.

The validity of Kenya's National Tallying Centre is highly doubted and questioned as evidenced by perennial legal challenges. For example, Senator Okiya Omtatah argued it introduces unnecessary delays and risks manipulation, pointing to past election disputes in 2013, 2017, and 2022 as evidence of systemic flaws (Wangui, 2025). The 2017 national elections witnessed violent protest erupt following a claim after the late Rt. Hon. Raila Amollo Odinga, opposition leader at the time, alleging database infiltration from hacker. Allegedly, the hackers had gained access to the database using the identity of the late Mr. Christopher Msando who was found dead after being tortured days before the election (BBC, 2017). Eventually, the Supreme Court of Kenya nullified the elections, finding that the electoral process and constitution were not properly followed. Such recurring fraud allegations erode public confidence in electoral outcomes. Despite widespread skepticism about electoral integrity, a majority of Kenyans still turn out to vote, creating a paradox where citizens participate in a process they do not fully trust. Thus, this trust deficit undermines the legitimacy of elected governments and fuels political instability.

The 2010 constitution of Kenya requires that after counting votes in polling stations, the IEBC shall tally and verify the counts then proceed to declare the results. This way, the constitution implicit the requirement of technological systems to aid in efficiency and transparency. Moreover, court rulings have established results certified by a returning officer at a constitutional level are final and can only be challenged by a petition in the Supreme court through an election petition (Kenya Law Blog, 2017). As a result, tension is created between the need for centralized verification and constitutional finality. Current electronic systems lack end-to-end verifiability, offer no protection against voter coercion, and fail to provide the paper audit trail required by Kenyan law for manual recounts. The 2022 election saw improved biometric verification using ten-finger capture and facial recognition, yet the system still faced scrutiny over result transmission and tallying procedures.

Therefore, Kenya needs an electoral system that simultaneously achieves five critical objectives: (1) *prevents and detects coercion*, (2) *provides mathematical proof of electoral integrity*, (3) *maintains complete voter privacy*, (4*) enables independent verification by all stakeholders*, and (5) *complies with constitutional requirements for paper audit trails*. No existing solution addresses all five requirements. Without a trustworthy, verifiable, and coercion-resistant voting system, Kenya faces continued cycles of electoral violence, disputed results, and weakened democratic institutions. Historically, such patterns have already cost thousands of lives and billions in economic damage.

Proposed Innovative Solution

Verivote represents a first of its kind hybrid electoral voting system combining trustworthiness of paper ballots with cutting-edge cryptography and artificial intelligence to create a “mathematically provable democracy.” Unlike most traditional and some contemporary systems which require blind trust in technology or officials, VeriVote enables every stakeholder—from individual voters to international observers—to independently verify electoral integrity while maintaining absolute ballot secrecy.

The system’s architecture integrates 5 revolutionary technologies: Soul-Bound Tokens (SBTs) for tamper-proof digital identity, Zero-Knowledge Proofs (ZKPs) for verification without revealing secrets, distributed mixnets for anonymization, homomorphic encryption for privacy-preserving tallying, and AI-powered security monitoring for real-time threat detection. This way, Verivote restores electoral legitimacy while addressing Kenya’s constitutional requirements.

Solving Existing Electoral Challenges

Digital Identity with SBTs

At the foundation of VeriVote's security architecture lies Soul-Bound Token (SBT) technology—non-transferable digital credentials that cryptographically bind voting rights to individual citizens. Unlike traditional voter registration cards that can be forged, stolen, or duplicated, SBTs leverage blockchain technology to create digital identities that are mathematically impossible to transfer or replicate (Pinna et al., 2024). This technology has been successfully adopted by Binance in its Binance Account Bound (BAB) tokens to provide on-chain verifiable identity confirming users have completed "Know-Your-Customer" (KYC) preventing fraud. This way, each registered Kenyan voter receives an SBT linked to their national identification and biometric data during the registration period. This eliminates ghost voters, duplicate registrations, and identity fraud that have plagued previous Kenyan elections.

Voter Coercion

VeriVote's anti-coercion strategy operates on three levels including cryptographic safeguards, behavioral monitoring through artificial intelligence, and procedural flexibility. Each voter possesses two Personal Identification Numbers linked to their SBT- a normal PIN for standard voting and a distress PIN that silently signals coercion. When a voter under duress enters their distress PIN at a shared voting device, the system appears to accept their vote normally—satisfying any coercer watching—but simultaneously flags their account through encrypted channels and notifies election authorities. This is strengthened by VeriVote's multiple-voting capability, allowing voters to change their vote throughout voting period. Only the voter's final submission counts. To prevent last-minute manipulation attacks, the system uses a time-locked protection- unlimited casting of votes in the first 12hrs, in-person verification during the next 8hrs, and complete freeze in the last 2hrs. To further complement Verivote, a machine learning system trained to detect patterns indicative of systematic coercion, vote buying, coordinated fraud, and electoral manipulation is implemented. The AI continuously analyzes voting patterns across temporal, geographic, and demographic dimensions, flagging anomalies such as sudden voting surges from specific regions, statistically improbable vote distributions, coordinated use of distress PINs suggesting organized intimidation campaigns, and voting patterns consistent with predetermined outcomes. When the AI assigns a suspicion score >=70% election officials

conduct in-person verification. However, the AI does not block users from voting or reject voting but logs the activity on the blockchain.

Establishing Mathematical Proof of Electoral Integrity

VeriVote addresses Kenya's persistent fraud allegations by replacing trust-based systems with mathematical certainty. When a voter makes selections on a shared voting device at their polling station, the system encrypts their vote using homomorphic encryption and generates a QR code containing the encrypted vote, a Zero-Knowledge Proof, a unique serial number, and cryptographic signatures. The voter immediately scans this QR code with their personal smartphone using the VeriVote verification app, which performs local cryptographic verification. This dual-device verification model prevents compromised voting terminals from manipulating results, as voters would immediately detect discrepancies on their independent personal devices. Once verified, the encrypted vote is recorded on a permissioned blockchain distributed across independent nodes operated by the IEBC, opposition party, international observers, civil society organizations, and universities. The blockchain requires >=4 of the 7 nodes to validate each transaction, ensuring no single party can manipulate the ledger. Critically, the blockchain stores only encrypted hashes-not actual vote contents- ensuring that even blockchain validators cannot see how anyone voted.

VeriVote deliberately employs homomorphic encryption rather than AI for the critical task of vote tallying, and this design choice reflects fundamental principles of electoral integrity. While AI excels at pattern recognition and anomaly detection- making it ideal for security monitoring- it is fundamentally unsuitable for vote counting because AI systems are probabilistic, not deterministic. An AI model might achieve 99.9% accuracy, but even 0.1% error across 30 million votes means 30,000 miscounted ballots—an unacceptable margin that would invalidate any election. In contrast, homomorphic encryption, is mathematically deterministic and provides cryptographic guarantees of correctness. It allows encrypted votes to be summed without ever decrypting individual ballots, producing a final tally that is mathematically proven to be accurate. The encryption scheme ensures that any attempt to manipulate vote totals would require breaking cryptographic assumptions that are computationally infeasible even for nation-state adversaries. Furthermore, homomorphic tallying produces verifiable cryptographic proofs that anyone with mathematical knowledge can independently verify- creating transparency that AI's "black box" decision-making cannot provide. In an electoral context where disputes must be

adjudicated in court, cryptographic proofs are legally defensible in ways that AI predictions are not.

Privacy Preservation Through Distributed Anonymization

Upon closure of voting, all encrypted votes enter a distributed mixnet—a network series of >=2 independent servers that shuffle and re-encrypt votes multiple times. Each server provides Zero-Knowledge Proofs demonstrating correct shuffling without revealing the permutation. This process mathematically guarantees that even if an observer knew exactly when a voter cast their ballot, they could not identify that vote in the final shuffled collection, breaking any connection between voter identity and vote choice. The anonymized votes are then tallied using homomorphic encryption, producing encrypted aggregate totals. Only when the final tally is ready do the7 IEBC commissioners physically gather for a live-streamed decryption ceremony. The system employs threshold cryptography, requiring five of the seven commissioners to simultaneously provide their decryption key shares. This design prevents any individual commissioner from accessing results prematurely or manipulating the tally, while ensuring that even if two commissioners are compromised, incapacitated, or act maliciously, the remaining five can still decrypt the final count.

Cryptographically-Bound Paper Audit Trails

To comply with Kenya's constitutional requirements for physical ballots, VeriVote implements post-election mass paper printing. After the final encrypted tally is verified, the system prints paper ballots for all valid votes at a centralized secure facility under multi-party observation by all seven IEBC commissioners, party agents, and international observers. Each printed ballot contains human-readable selections, a QR code with the encrypted vote and blockchain Merkle proof, and cryptographic signatures from all observing parties. These paper ballots are cryptographically bound to the blockchain, meaning anyone can scan a ballot’s QR code and verify it corresponds exactly to a blockchain-recorded vote. This bidirectional verification creates a system where digital records prove paper integrity while paper provides legal backup for manual recounts. Therefore, this satisfies the Supreme Court’s requirements for verifiable, auditable elections while reducing paper costs by 90% compared to traditional voting. Thus, VeriVote creates a system where democracy can be proven, not merely proclaimed.

Technology and Methodology

1. Blockchain & Distributed Ledger Technology

Hyperledger Fabric v2.5+ (Primary Blockchain Platform)

We selected Hyperledger Fabric as our permissioned blockchain foundation for several strategic reasons. Unlike public blockchains that sacrifice transaction throughput for decentralization, Hyperledger Fabric achieves 3,500+ transactions per second while maintaining enterprise-grade security through its modular architecture. The platform's channel-based privacy model allows us to segregate different electoral processes—voter registration, vote casting, and tallying—into separate channels with distinct access controls, preventing unauthorized data exposure.

Polygon (Ethereum Layer 2) - Alternative Implementation

For jurisdictions preferring public blockchain transparency or international interoperability, we maintain a parallel implementation on Polygon, Ethereum's leading Layer 2 scaling solution. Polygon provides the transparency benefits of public blockchains—anyone globally can run a full node and verify election integrity—while achieving 7,000+ transactions per second at costs below $0.01 per transaction through its Proof-of-Stake sidechain architecture.

Smart contracts deployed on Polygon use Solidity v0.8.20+ with OpenZeppelin's audited contract libraries for standardized implementations of access control, cryptographic primitives, and upgrade mechanisms. The ERC 1155 multi-token standard efficiently manages Soul-Bound Tokens for voter identity while minimizing gas costs. Polygon's compatibility with Ethereum tooling (Hardhat, Ethers.js, MetaMask) accelerates development and enables integration with the broader Web3 ecosystem.

InterPlanetary File System (IPFS) - Decentralized Storage

Large data objects—electoral regulations, candidate manifestos, audit logs, and post-election paper ballot images—are stored on IPFS rather than directly on-chain, with content-addressed hashes recorded in the blockchain. This hybrid approach maintains blockchain immutability while avoiding prohibitive storage costs. Pinata provides redundant IPFS pinning across geographically distributed nodes, ensuring content availability even if individual nodes fail. Content-addressed storage provides automatic deduplication and cryptographic verification that retrieved data matches recorded hashes.

2. Cryptographic Frameworks

Microsoft SEAL v4.1+ (Homomorphic Encryption)

Microsoft SEAL (Simple Encrypted Arithmetic Library) implements the BFV (Brakerski-Fan-Vercauteren) and CKKS homomorphic encryption schemes, enabling arithmetic operations on encrypted vote tallies without decryption. We selected SEAL over alternatives like HElib or PALISADE due to its superior documentation, active Microsoft Research backing, and optimized performance on commodity hardware. SEAL's security parameters are configured to provide 128-bit post-quantum security, ensuring resistance to future quantum computing attacks.

Circom v2.1+ & SnarkJS (Zero-Knowledge Proofs) enable voters to prove their encrypted ballot is valid—properly formatted, containing exactly one selection per race, and conforming to electoral rules—without revealing their choices. We implement ZKPs using Circom, a domain-specific language for defining arithmetic circuits, compiled to zk-SNARK (Zero-Knowledge Succinct Non-Interactive Argument of Knowledge) proofs via the Groth16 proving system.

Mixnet Implementation (Chaumian Mixing with Verificable Shuffles)

Vote anonymization occurs through a distributed mixnet architecture inspired by David Chaum's pioneering work. Five or more independent servers operated by mutually distrusting organizations each perform verified shuffles using the Neff shuffle proof system, which provides zero-knowledge proofs that output votes are permutations of input votes without revealing the permutation itself.

Libsodium (General Cryptographic Primitives)

Libsodium provides battle-tested implementations of cryptographic primitives for routine operations: Ed25519 signatures for authentication, XSalsa20-Poly1305 authenticated encryption for data protection, Argon2 for password hashing (voter PIN derivation), and X25519 for key exchange. We prioritize libsodium over OpenSSL due to its secure-by-default API design that prevents common implementation mistakes like nonce reuse or forgetting authentication tags.

3. Artificial Intelligence & Machine Learning

Anomaly Detection: Scikit-learn v1.4+

We employ Scikit-learn's Isolation Forest for real-time voting pattern anomaly detection. This unsupervised learning algorithm excels at identifying outliers without requiring labeled fraud examples—a critical advantage in electoral contexts where fraud patterns evolve and historical labeled datasets are scarce. The algorithm works by randomly partitioning data into isolation

trees; anomalies require fewer partitions to isolate than normal observations, providing an efficient detection mechanism.

The model ingests real-time features extracted from voting activity:

- Voting velocity: votes per 5-minute window per polling station
- Temporal patterns: hourly voting distribution compared to historical baselines
- Capacity ratios: votes received versus physical station capacity
- Geographic clustering: correlation of patterns across nearby stations
- Device patterns: voting sequences from specific shared devices

Training occurs on synthetic datasets generated from publicly available Kenyan electoral statistics (2013, 2017, 2022 elections) augmented with simulated fraud scenarios including ballot stuffing (sudden vote surges), coordinated manipulation (multiple stations showing identical suspicious patterns), and machine voting (statistically improbable timing uniformity). The model assigns anomaly scores from 0-100, with scores above 70 triggering mandatory human review by election officials.

Intelligent Pattern Analysis: OpenAI API Integration

Cloud-Based Natural Language Intelligence

For complex pattern interpretation requiring contextual reasoning, we leverage OpenAI's GPT-4 API (specifically the cost-optimized GPT-4o-mini model at $0.150 per million input tokens, $0.600 per million output tokens). This integration enables sophisticated natural language analysis without investing months in custom model development. When the Isolation Forest flags suspicious patterns, GPT-4 analyzes feature vectors and generates human-readable explanations. For instance, “Station X shows voting rate 40% above capacity during 14:00-15:00 window, coinciding with similar patterns at 5 nearby stations—suggests coordinated manipulation.” Additionally, the API synthesizes multiple weak signals into coherent threat assessments. For instance, analyzing whether geographic clustering of anomalies aligns with known ethnic tensions, recent incidents, or social media chatter. Finally, it provides election officials with recommended actions based on pattern severity and historical precedent.

Coercion Pattern Detection: Hidden Markov Models

For detecting systematic coercion campaigns through distress PIN patterns, we employ Hidden Markov Models (HMMs) from the hmmlearn library. HMMs model the probability distribution of distress PIN usage across time, geography, and demographics, distinguishing

isolated coercion incidents (expected, handled through revote protocols) from organized intimidation campaigns requiring intervention.

4. Application Development Frameworks

Backend: Node.js v20+ with TypeScript v5+

Node.js provides our backend runtime, selected for its non-blocking I/O model ideal for handling thousands of concurrent verification requests during peak voting hours. TypeScript adds static typing, reducing runtime errors and improving code maintainability for our distributed development team. The Express.js v4.19+ framework handles HTTP routing, with Fastify v4.26+ providing high-performance alternatives for latency-sensitive endpoints.

Mobile: React Native v0.74+ with Expo v51+

The voter verification app uses React Native, enabling simultaneous deployment to iOS and Android from a single TypeScript codebase. Expo simplifies build processes and provides over-the-air updates, allowing critical security patches without requiring users to update through app stores—essential for rapidly responding to discovered vulnerabilities.

5. Database & Data Management

PostgreSQL v16+ (Primary Relational Database)

PostgreSQL stores non-blockchain data including voter registration details (linked to SBT addresses but not to votes), polling station configurations, device attestation records, and audit logs. We leverage PostgreSQL's JSONB datatype for flexible schema evolution, full-text search for audit log queries, and row-level security policies enforcing principle of least privilege.

Redis v7+ (Caching & Real-Time Operations)

Redis provides sub-millisecond response times for frequently accessed data: voter SBT lookups, rate limiting counters (preventing DoS attacks), session management, and real-time vote count aggregations displayed on the public portal. Redis Streams enable event-driven architectures, publishing vote recording events to AI analytics services and monitoring dashboards

TimescaleDB (Time-Series Analytics)

TimescaleDB, a PostgreSQL extension optimized for time-series data, stores voting pattern metrics at 1-second granularity. The AI analytics service queries TimescaleDB for features like: votes per minute per station over the past hour, inter-vote timing distributions, and cumulative

turnout curves. Automatic data retention policies compress older data while maintaining query performance.

6. Cloud Infrastructure & DevOps

Amazon Web Services (Primary Cloud Provider)

EC2 instances run blockchain peer nodes (c6i.2xlarge for compute-intensive consensus), API servers (t3.large auto-scaling groups), and AI inference (g5.xlarge with NVIDIA T4 GPUs for real-time anomaly detection). Moreover, AWS Managed Blockchain provides hosted Hyperledger Fabric with automated certificate authority, ordering service management, and monitoring. Additionally, application Load Balancers distribute traffic across availability zones, with Route 53 providing DNS-based geographic routing and health checks. VPNs secure communication between IEBC-controlled infrastructure and cloud resources. Finally, with KMS (Key Management Service), encryption keys with hardware security modules. IAM enforces principle of least privilege with role-based access control.

Kubernetes v1.29+ (Container Orchestration)

Kubernetes orchestrates containerized microservices, providing automatic scaling, rolling updates, and self-healing. Helm charts define application deployments, enabling consistent deployments across development, staging, and production environments. The service mesh (Istio v1.21+) provides mutual TLS between services, distributed tracing, and circuit breaking for resilience.

Relevance to Theme: AI for National Prosperity

VeriVote Kenya embodies the transformative potential of artificial intelligence to strengthen the foundational pillar of national prosperity: legitimate democratic governance. Electoral integrity directly determines whether nations experience sustainable development or descend into cycles of violence and economic stagnation. Kenya's history starkly illustrates this connection—the 2007-2008 post-election violence cost over 1,100 lives and erased $3.6 billion in economic output, demonstrating how disputed elections devastate national prosperity. VeriVote deploys AI not merely as technological enhancement but as a strategic intervention addressing Kenya's most critical governance challenge.

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